

MS 6015: Multivariate Statistical Methods for Business

Multivariate Inference about Means*

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**Disclaimer: These outline notes are to help the student understand a few main concepts. They have not been proof-read for technical errors. Students are advised to validate the results using recommended textbooks.*

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1 Univariate Inference

Review of Notation

	Population	Sample
Mean	μ	$\bar{X}$
Standard Deviation	σ	s
Proportion	p	$\hat{p}$

1.1 Sampling distribution

- A sampling distribution is created by, as the name suggests, sampling.
- Two methods
 - Actually draw samples of the same size from a population, calculate the statistic of interest, and then use descriptive techniques to learn more about sampling distribution.
 - Rely on the rules of probability and the laws of expected value and variance to derive the sampling distribution.

Central Limit Theorem

- The *sampling distribution* of the mean of a random sample drawn from any population is *approximately normal* for a sufficiently large sample size.
 - The larger the sample size, the more closely the sampling distribution of $\bar{X}$ will resemble a normal distribution.
- If the population is normal, then $\bar{X}$ is normally distributed for all values of n .
- If the population is non-normal, then $\bar{X}$ is approximately normal only for larger values of n .
- In most practical situations, a sample size of 30 may be sufficiently large to allow us to use the normal distribution as an approximation for the sampling distribution of $\bar{X}$.

Sampling Distribution of the Sample Mean

- $\mu_{\bar{X}} = \mu$

- $\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$ and $\sigma_{\bar{X}} = \sigma/\sqrt{n}$. The standard deviation of the sampling distribution is called the **standard error** of the mean
- If X is normal, $\bar{X}$ is normal. If X is nonnormal, $\bar{X}$ is approximately normal for sufficiently large sample sizes.
- We can express the sampling distribution of the mean as

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

- The summaries above assume that the population is infinitely large. However if the population is finite¹ the standard error is

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

N is the population size and $\sqrt{\frac{N-n}{N-1}}$ is called the finite population correction factor.

1.2 Confidence intervals

- The $100(1 - \alpha)\%$ confidence interval for μ is

$$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

- When the population standard deviation is unknown, it must be replaced by the sample statistic
- This yields the summary statistic

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

- If a random sample is taken from a normal population, then the statistic has a t distribution with d.f. = $n - 1$

¹As a rule of thumb, we will treat any population that is at least 20 times larger than the sample size as large

- As sample size increases, the t -distribution approaches the z -distribution
- The $100(1 - \alpha)\%$ confidence interval for μ is

$$\left[\bar{X} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \bar{X} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \right]$$

- The value of t depends on the level of confidence and $n - 1$ degrees of freedom.

1.3 Hypothesis testing

- Hypothesis testing is the second form of statistical inference. It also has greater applicability.
- There are two hypotheses. One is called the null hypothesis and the other the alternative or research hypothesis. The usual notation is:
 - H_0 : the ‘null’ hypothesis
 - H_1 : the ‘alternative’ hypothesis
- Once the null and alternative hypotheses are stated, the next step is to randomly sample the population and calculate a **test statistic**.
- If the test statistic’s value is inconsistent with the null hypothesis we reject the null hypothesis and *infer that the alternative hypothesis is true*
- There are two possible decisions:
 - Conclude that there is enough evidence to support the alternative hypothesis.
 - Conclude that there is not enough evidence to support the alternative hypothesis.
- Two possible errors can be made.
 - Type I error: Reject a true null hypothesis; $P(\text{Type I error}) = \alpha$
 - Type II error: Do not reject a false null hypothesis; $P(\text{Type II error}) = \beta$

Testing for Means when σ is known

When the population standard deviation is known and the population is normal, the test statistic for testing hypotheses about μ is:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Testing for Means when σ is not known

When the population standard deviation is unknown and the population is normal, the test statistic for testing hypotheses about μ is:

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

which is Student t distributed with $n - 1$ degrees of freedom.

p -value: The p -value approach (which is generally used with a computer and statistical software).

- The p -value of a test is the probability of observing a test statistic at least as extreme as the one computed given that the null hypothesis is true.
- Interpret the p -value as a weight of evidence against H_0 or as the plausibility of H_0 .
 - Small p -values mean that H_0 is not plausible.

Parts of a Hypothesis Test

1. State Null and alternative Hypothesis
2. Collect Data
3. Calculate the Test Statistic
4. Compute p -value
5. Interpret the results

Summary of Univariate Inference of Means

- Is a given value μ_0 a plausible value for the population mean μ ?
- We formulate the problem as a hypothesis testing problem. The competing hypotheses are

$$H_0 : \mu = \mu_0 \text{ and } H_1 : \mu \neq \mu_0$$

- Given a sample $X_1, X_2; \dots, X_n$ from a normal population, we compute the **test statistic**

$$t = \frac{(\bar{X} - \mu_0)}{s/\sqrt{n}}$$

where

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \text{ and } s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- If t is "small", then $\bar{X}$ and μ_0 are close and we fail to reject H_0
- When H_0 is true, the statistic t has a student t distribution with $n-1$ degrees of freedom.
- We reject the null hypothesis at level α when $|t| > t_{(n-1)}(\alpha/2)$
- Notice that rejecting H_0 when t is large is equivalent to rejecting it when the squared standardized distance

$$t^2 = \frac{(\bar{X} - \mu_0)^2}{s^2/n} = n(\bar{X} - \mu_0)(s^2)^{-1}(\bar{X} - \mu_0)$$

is large.

- We reject H_0 when

$$n(\bar{X} - \mu_0)(s^2)^{-1}(\bar{X} - \mu_0) > t_{n-1}^2(\alpha/2)$$

Confidence Intervals

- If we fail to reject H_0 , we conclude that μ_0 is close (in units of standard deviations of $\bar{X}$) to $\bar{X}$, and thus is a plausible value for μ .
- The set of plausible values for μ is the set of all values that lie in the $100(1 - \alpha)\%$ confidence interval for μ :

$$\left[\bar{X} - t_{(n-1)}(\alpha/2) \frac{s}{\sqrt{n}}, \bar{X} + t_{(n-1)}(\alpha/2) \frac{s}{\sqrt{n}} \right]$$

- The confidence interval consists of all the values that would not be rejected by the α level test of $H_0 : \mu = \mu_0$.

1.4 Comparing means of two populations

1.4.1 Independent samples

Sampling Distribution of $\bar{X}_1 - \bar{X}_2$

- $\bar{X}_1 - \bar{X}_2$ is normally distributed if the original populations are normal or approximately normal if the populations are nonnormal and the sample sizes are large ($n_1, n_2 > 30$)
- The expected value of $\bar{X}_1 - \bar{X}_2$ is $\mu_1 - \mu_2$
- The variance of $\bar{X}_1 - \bar{X}_2$ is $\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$
- The standard error is $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
- $Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$

In practice, the Z statistic is rarely used since the population variances are unknown. We consider two cases $\sigma_1^2 = \sigma_2^2$ and $\sigma_1^2 \neq \sigma_2^2$

Making Population Inferences with Unknown Population Variances.

Case 1: $\sigma_1^2 = \sigma_2^2$:

- Calculate the pooled variance estimate

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

and use the t - statistic defined by

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

with $\nu = n_1 + n_2 - 2$ degrees of freedom.

- The confidence interval estimator for $\mu_1 - \mu_2$ when the population variances are equal is given by:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

with ν degrees of freedom

Making Population Inferences with Unknown Population Variances. Case 2: $\sigma_1^2 \neq \sigma_2^2$:

- The t - statistic defined by

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

with $\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$ degrees of freedom.

- The confidence interval estimator for $\mu_1 - \mu_2$ when the population variances are unequal is given by:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

with ν degrees of freedom

1.4.2 Paired Comparisons

- By design, the n differences

$$D_j = X_{j1} - X_{j2}$$

- Sampling distribution. If $D_j \sim N(\delta, \sigma_D^2)$, the variable

$$t = \frac{\bar{D} - \delta}{s_D / \sqrt{n}} \sim \text{Student's } t_{n-1} \text{ - distribution}$$

$$\text{where } \bar{D} = \frac{1}{n} \sum_{j=1}^n D_j \text{ and } s_D^2 = \frac{1}{n-1} \sum_{j=1}^n (D_j - \bar{D})^2$$

- Hypothesis Testing

$$H_0 : \delta = 0$$

$$H_1 : \delta \neq 0$$

- A $100(1 - \alpha)\%$ confidence interval estimate of δ is given by

$$\bar{D} \pm t_{n-1}(\alpha/2) \frac{s_D}{\sqrt{n}}$$

1.5 Anova: Analysis of Variance

One-Way Analysis of Variance

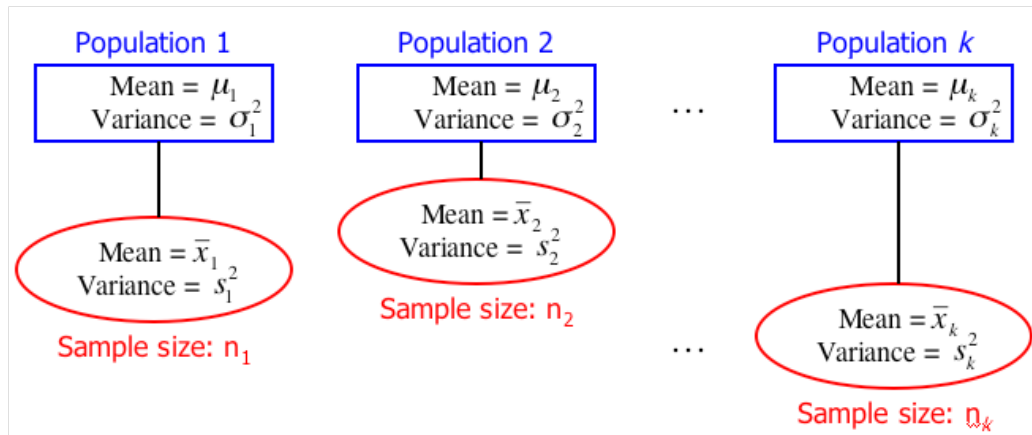
- Independent samples are drawn from k populations: [These populations are referred to as treatments](#)

It is not a requirement that $n_1 = n_2 = \dots = n_k$

Data Table

		Treatment			
		1	2	...	k
Subject	1	X_{11}	X_{21}	...	X_{k1}
	2	X_{12}	X_{22}		X_{k2}
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	n_i	X_{1n_1}	X_{2n_2}	...	X_{kn_k}

Terminology



- X is the response variable, and its values are responses.
- X_{lj} refers to the j^{th} observation in the l^{th} treatment. E.g. X_{35} is the fifth observation of the third treatment.
- The grand mean, $\bar{X}$, is the mean of all the observations, i.e.:

$$\bar{X} = \frac{\sum_{l=1}^k \sum_{j=1}^{n_l} X_{lj}}{n}; n = n_1 + n_2 + \dots + n_k$$

- Population classification criterion is called a **factor**.
- Each population is a **factor level**.

One-Way ANOVA The univariate case where $p = 1$

- Assumptions: $X_{lj} \sim N(\mu_l, \sigma^2)$ i.i.d FOR $j = 1, \dots, n_l$ and $l = 1, 2, \dots, k$
- Hypothesis: $H_o : \mu_1 = \mu_2 = \dots = \mu_k$ versus H_1 : not H_o

$$\begin{aligned} \underbrace{\mu_l}_{l^{th} \text{ population mean}} &= \underbrace{\mu}_{\text{grand mean}} + \underbrace{\mu_l - \mu}_{l^{th} \text{ treatment effect}} \\ &= \mu + \tau_l \end{aligned}$$

- If $\mu_1 = \mu_2 = \dots = \mu_k$, then an equivalent way to write the null hypothesis is

$$H_o : \tau_1 = \tau_2 = \dots = \tau_k = 0$$

The Model for an Observation

$$X_{lj} = \mu + \tau_j + \epsilon_{lj}$$

where $\epsilon_{lj} \sim N(0, \sigma^2)$ and independent.

- ϵ_{lj} is **random error**

- **Identification constraint:** $\sum_{l=1}^k \tau_l = 0$

- Decomposition of an observation is

$$\underbrace{X_{lj}}_{\text{observation}} = \underbrace{\bar{X}}_{\substack{\text{overall sample} \\ \text{mean}}} + \underbrace{(\bar{X}_l - \bar{X})}_{\substack{\text{estimated} \\ \text{treatment} \\ \text{effect}}} + \underbrace{(X_{lj} - \bar{X}_l)}_{\substack{\text{residual} \\ \text{error}}}$$

$\bar{X}$ is the estimator of μ ; $\hat{\tau}_l = (\bar{X}_l - \bar{X})$ is estimator of τ_l and $(X_{lj} - \bar{X}_l)$ is estimator of ϵ_{lj}

The Sum of Squares

- Sum of Squared Observations

$$SS_{obs} = SS_{total} = \sum_{l=1}^k \sum_{j=1}^{n_l} X_{lj}^2$$

- The three components of X_{lj} can be taken and sum of squares can be formed as

$$- SS_{mean} = \sum_{l=1}^k \sum_{j=1}^{n_l} \bar{X}^2 = \left(\sum_{l=1}^k n_l \right) \bar{X}^2$$

$$- SS_{treatment} = \sum_{l=1}^k \sum_{j=1}^{n_l} \hat{\tau}_l^2 = \sum_{l=1}^k \sum_{j=1}^{n_l} (\bar{X}_l - \bar{X})^2 = \sum_{l=1}^k n_l (\bar{X}_l - \bar{X})^2$$

$$- SS_{residual} = \sum_{l=1}^k \sum_{j=1}^{n_l} \hat{\epsilon}_{ij}^2 = \sum_{l=1}^k \sum_{j=1}^{n_l} (X_{lj} - \bar{X}_l)^2$$

Test Statistic

- Since $\mu_1 = \mu_2 = \mu_3 = \mu_4$ is of interest to us, a statistic that measures the **proximity of the sample means to each other** would also be of interest.
- Such a statistic exists, and is called the **between-treatments variation**. It is denoted SST, short for “sum of squares for treatments”. Its is calculated as:

$$SS_{treatment} = \sum_{l=1}^k n_l (\bar{X}_l - \bar{X})^2$$

- A large SST indicates large variation between sample means which supports H_1 .
- The mean square for treatments (MST) is given by:

$$MST = \frac{SST}{k-1}$$
- The mean square for errors (MSE) is given by:

$$MSE = \frac{SSE}{n-k}$$
- The Test statistic:

$$F = \frac{MST}{MSE}$$

is F -distributed with $k - 1$ and $n - k$ degrees of freedom.

- $p - value = P(F > F_{stat})$
 - The usual F -test rejects H_0 at level α is $F = \frac{MST}{MSE} > F_{k-1, n-k}(\alpha)$
 - Equivalently, reject H_0 for large values of $\frac{SST}{SSE}$ or for large values of $1 + \frac{SST}{SSE}$
 - The statistic appropriate for a multivariate generalization rejects H_0 for small values of

$$\frac{1}{1 + \frac{SST}{SSE}} = \frac{SSE}{SST + SSE}$$